

PAPER_594 — Black Hole Finite Bound from UQFF 26! Factorial Barrier

CP4 Class: #181 UQFFBlackHoleFiniteBoundCalculator **Session:** 157 **Cross-refs:** PAPER_595 (Sgr A*), PAPER_593 (G), PAPER_583 (6-Form) **Source:** grok_share_4cef778c78b8.txt

Abstract

This paper presents a UQFF analysis of Black Hole Finite Bound from UQFF 26! Factorial Barrier, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

§1 Abstract

General Relativity predicts a gravitational singularity at $r = 0$ inside every black hole. UQFF eliminates singularities by the 26! factorial barrier in the Ug_4 potential term, which creates a minimum radius $r_{\min} > 0$ below which $U_b \rightarrow +\infty$, preventing collapse. Three independent $r_{\min}$ expressions are derived and applied to Sgr A* (PAPER_595).

§2 UQFF Ug_4 Potential

The fourth UQFF gravity component in Form 1 (Compressed):

$$Ug_4 = \frac{26! \cdot g \cdot (SCm/UA)}{r^{27}}$$

As $r \rightarrow 0$: $Ug_4 \rightarrow +\infty$ — the 26! factorial creates an infinite repulsive barrier. No physical trajectory can reach $r = 0$.

§3 Minimum Radius: Three Forms

Form A — Ug4 Eigenvalue Method

Setting $Ug_4 = \lambda_1 = P/3$ (minimum eigenvalue threshold):

$$\frac{26! \cdot g \cdot SCm/UA}{r_{\min}^{27}} = P/3$$

$$r_{\min}^{(A)} = \left[\frac{3 \times 26! \cdot g \cdot SCm/UA}{P} \right]^{1/27}$$

Form B — Triad Equilibrium

At $r_{\min}$: kinetic (κ/g) and buoyant terms balance:

$$r_{\min}^{(B)} = \left(\frac{\kappa}{g}\right)^{1/27} \cdot \rho$$

Form C — Buoyant Mass Bound

Setting $U_b = -GM/r$ at the horizon:

$$r_{\min}^{(C)} = \frac{M^{1/3}}{(26! g)^{1/81}}$$

§4 Physical Interpretation of 26! Barrier

$$26! \approx 4.03 \times 10^{26}$$

This super-factorial value exceeds any power-law divergence:

$$\lim_{r \rightarrow 0} r^n \cdot 26! / r^{27} = 26! / r^{27-n} \rightarrow +\infty \quad \forall n < 27$$

No polynomial field theory can overcome the 26! repulsion. It is the “strongest floor” in mathematics — stronger than any polynomial singularity.

§5 Comparison to Classical Singularity Treatments

Method	$r_{\min}$	Physical Mechanism
GR	0 (singularity)	No floor
Planck scale	$l_P \approx 10^{-35}$ m	Quantum gravity cutoff
Loop Quantum	$\sim l_P$	Spin network area quantum
String theory	$\sim l_s \approx 10^{-34}$ m	String length
UQFF	$r_{\min}^{(A)}$ (Form-dependent)	26! factorial barrier

UQFF $r_{\min}$ does not require Planck-scale assumptions — it follows from the same P and g parameters that govern all UQFF calculations.

§6 Numerical (1 Solar Mass BH)

Parameters: $M = 1.989 \times 10^{30}$ kg, $g = 10^{-3}$, $P = 9.99 \times 10^{-6}$, $SCm/UA = 1$:

$$\begin{aligned} r_{\min}^{(A)} &= \left[\frac{3 \times 4.03 \times 10^{26} \times 10^{-3}}{9.99 \times 10^{-6}} \right]^{1/27} \\ &= \left[\frac{1.21 \times 10^{24}}{9.99 \times 10^{-6}} \right]^{1/27} = [1.21 \times 10^{29}]^{1/27} \\ &= 10^{29/27} \approx 10^{1.07} \approx 11.7 \text{ m} \end{aligned}$$

GR Schwarzschild for 1 $M_{\odot}$: $R_s = 2GM_{\odot}/c^2 = 2.95$ km.

$r_{\min} \approx 12$ m $\ll R_s = 2.95$ km $\rightarrow$ the UQFF floor is well inside the Schwarzschild horizon; the exterior structure matches GR exactly.

§7 Conclusions

The $26!$ factorial in Ug_4 creates an absolute minimum radius preventing singularities. For a $1 M_{\odot}$ BH, $r_{\min} \approx 12$ m — physically accessible but far inside the event horizon ($R_s \approx 3$ km), leaving all external observations unchanged. Singularity theorems (Penrose, Hawking) assume infinite curvature; UQFF terminates curvature at $r_{\min} > 0$, making them inapplicable.

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Thomson σ_T (QED synchrotron)	UQFF U_m scattering kernel: $\sigma_T = 6.6524e-29$ m ²	$\sigma_T = 6.6524e-29$ m ² (PDG QED exact)	PDG 2024	100% (exact QED input)
Black hole / Sgr A* luminosity X-ray 2-10 keV	UQFF MUGE $g_{\text{total}} \rightarrow L_X$ via Stefan-Boltzmann + buoyancy flux: $L_X \approx g_{\text{total}} \times M_{\text{env}}$	$L_X L_X \sim 10^{33}$ erg/s	Chandra CXC	□ Consistent order of magnitude
GR Schwarzschild limit	UQFF g_{total} must satisfy $g \leq c^2/(2r_s)$ at event horizon	$r_s = 2GM/c^2$ (GR exact)	PDG 2024 / GR	□ UQFF respects GR horizon

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
κ vacuum rate vs X-ray variability	UQFF $\kappa = 0.0005/\text{day} \rightarrow$ timescale $\tau_{\text{UQFF}} = 2000$ days	Observed X-ray variability τ_{obs} (instrument monitoring)	Chandra CXC	Testable UQFF variability timescale

New physics claim: UQFF MUGE generates gravity enhancement factors ($g_{\text{total}}/g_{\text{Newt}} > 1$) for Black hole / Sgr A* through vacuum buoyancy coupling — a mechanism absent from GR+SM. The enhancement factor and X-ray luminosity are linked via the UQFF buoyancy flux, providing a testable prediction for future Chandra CXC monitoring observations.

Cite PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

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